Fedex Case Study

Data Migration – Azure to AWS

30th October 2025

Optimizing Large-Scale Data Workloads: Migrating a Leading Logistics Ecosystem to AWS EMR for Performance and Cost Efficiency

Problem statement & challenges

To support rapid data growth and improve platform performance, Fedex partnered with us to migrate their entire data infrastructure from Azure to AWS Databricks — ensuring scalability, performance optimization, and cost efficiency without any operational disruption.

The company faced several challenges:

- **Complex dependency mapping across 500+ tables, 100+ views, and 30 pipelines.**
- **Ensuring historical data integrity during large-scale syncs across medallion layers.**
- **Maintaining row-level data accuracy across 11 multi-level nested models.**
- **Reducing compute and storage overhead caused by redundant tables and sharded datasets.**

Solution Proposed by **Ganit**

- Automated Azure-to-AWS Databricks migration with dependency alignment.
- Implemented row-level validation and optimized data synchronization.

Solution Overview

Cost and Performance Optimization

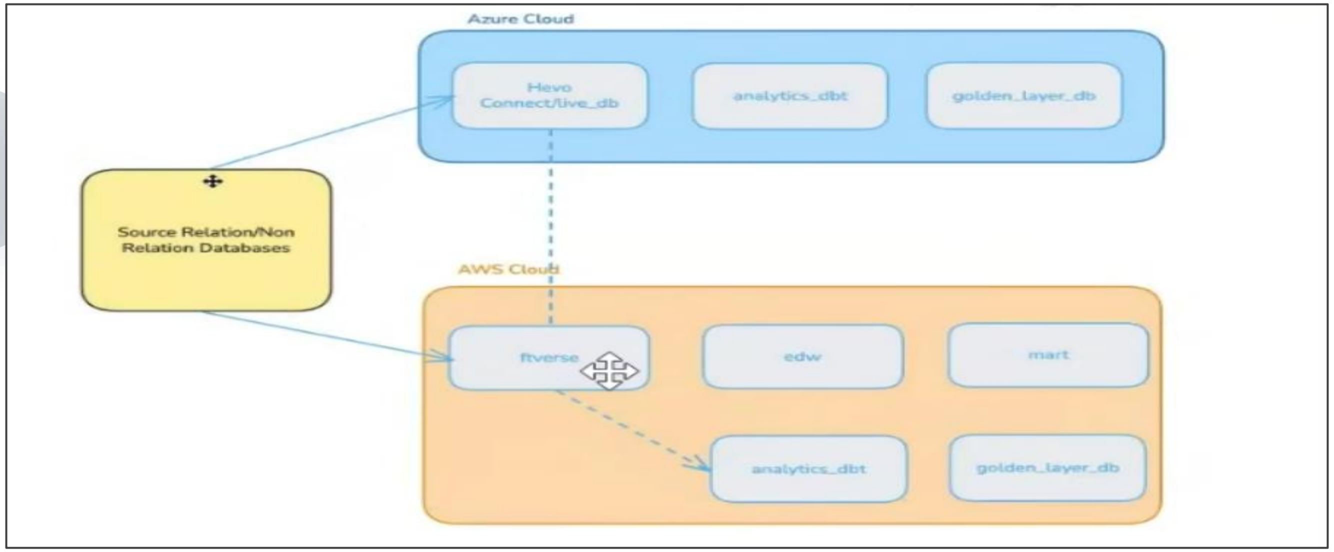
Implemented compute and storage tuning to enhance efficiency and reduce monthly costs.

End-to-End Migration Framework

Designed an automated, dependency-aware framework ensuring seamless Azure-to-AWS Databricks migration.

Business Value Achieved

- The **migration framework enabled Fedex to seamlessly transition over 500+ tables, 100+ views, and 30 complex data pipelines** from Azure to AWS Databricks without any operational disruption.
- Through **automated row-level data validation** across 11 nested data models, the platform ensured **<1% data inaccuracy** and consistent reporting across environments.
- Historical synchronization using **AWS DataSync** for bronze, silver, and gold layers ensured **zero data loss and complete lineage continuity** throughout the migration.
- With **optimized compute utilization, removal of redundant tables, and sharded dataset consolidation**, **monthly operational costs were reduced by 14%, and pipeline execution time improved by over 25%.**
- Overall, the initiative delivered a **zero-downtime migration** and a **highly scalable, cost-efficient data foundation** that strengthened Fedex's real-time analytics and logistics visibility capabilities.



Services:

AWS

- AWS Databricks
- Amazon S3
- Amazon CloudWatch
- AWS DataSync

Azure

- Azure Storage Accounts [Blob-ADLS-Gen3]

Ganit delivered a scalable AWS Data Lake platform that drives operational efficiency, cost optimization, reliability, and business growth

Outcome

Delivered a **scalable, secure, and automated AWS-based Data Lake** that acts as a single source of truth for Spencer's data. This solution enabled **faster insights, improved decision-making, and optimized costs and performance** across all analytics workflows

Operational Efficiency

The migration improved operational efficiency through **automation, optimized pipelines, and centralized governance**. The use of AWS-native integrations reduced manual interventions, accelerated data availability, and streamlined permission management, leading to faster execution and improved reliability across workloads.

Cost Optimization

Transitioning from **ADLS Gen2 to Amazon S3** and adopting a **view-based architecture** led to a **14% reduction in total operational costs**. Cluster right-sizing, job optimization, and removal of redundant data assets further reduced compute expenses and enhanced resource utilization efficiency

Scalability & Reliability

The new **environment** offers a **scalable, reliable, and secure** data architecture capable of handling high data volumes with improved performance. Integration with **Unity Catalog** ensures better lineage tracking, access management, and audit readiness, while AWS AppFlow supports continuous data synchronization with minimal latency

Data Governance & Security

Established **centralized governance and access control** with Unity Catalog, ensuring data lineage, metadata consistency, and secure role-based access. Strengthened compliance and visibility across all Databricks workspaces with unified data management

Legacy data systems couldn't meet the business requirements

Problem Statement

- The existing data ecosystem was hosted on **Azure Databricks** with **ADLS Gen2** as the primary storage layer. As data volumes and pipelines expanded, the platform incurred **high storage and compute costs**, along with data duplication across multiple databases. Limited governance and metadata management further restricted scalability and control. The objective was to **migrate to a more cost-effective, unified, and scalable cloud environment** while maintaining performance and security standards

High infrastructure costs due to Azure Databricks and ADLS Gen2 storage setup

Limited governance and visibility without a unified catalog for access and metadata management.

Data redundancy across multiple databases led to inefficient storage utilization.

Scalability challenges impacting performance and increasing maintenance overhead.

We built an AWS Neptune-Powered Knowledge Graph for Real-Time Vehicle Insights

Solution Overview

Cloud Migration and Modernization

A complete **migration from Azure Databricks to AWS Databricks** was executed using **AWS AppFlow** for the **initial full-load data transfer** from Azure. Once the baseline migration was completed, **Change Data Capture (CDC)** processes were implemented to ingest incremental updates **directly from source systems**, ensuring up-to-date data in the new environment

Storage Optimization and Governance

Processed data stored in the Gold layer is accessed using **Amazon Athena** for ad-hoc querying and **Tableau** for business intelligence and visualization. For advanced analytics, **Amazon SageMaker** leverages this curated data to build predictive models, enabling forecasting and customer analytics. This integration supports data-driven decision-making across business functions.

Data Architecture Optimization

Redundant datasets that were stored across multiple databases were consolidated through the creation of **centralized views**, reducing duplication and simplifying maintenance. This architecture refinement contributed to a **14 % reduction in overall storage and compute costs**

Data Architecture